

Learned Versus Pendulum Slosh Models in Sampling-Based MPC for Fast Spill-Free Cart Transport: A Small Simulation Study

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<https://github.com/mihir-chauhan/learned-slosh-mpc-fast-transport>

Abstract—A robot that carries a brim-full open container should move as fast as the liquid allows. Most slosh-aware planners predict the liquid with a fitted pendulum or spring-mass model; learned predictors are an alternative. We compare the two model classes inside one time-minimising sampling-based model predictive controller (MPPI), in a small CPU-only simulation: a planar cart moves a rectangular container 1 m, and the liquid is a reduced-order two-mode nonlinear slosh model chosen for this study, not a fluid simulation. Every controller gets one tuned scalar, its spill margin, selected by the same rule: the largest value with no spill on validation episodes. On the nominal liquid, MPC with a learned predictor that infers a latent liquid context from sensor history is spill-free at 1.99 s, against 2.15 s for the fitted pendulum and 2.04 s for the fitted single-mode spring-mass model (5 seeds). It is not the fastest system: a fitted two-mode spring-mass model reaches 1.96 s and open-loop zero-vibration input shaping 1.89 s. On held-out liquids with a lateral push, input shaping spills in 97 of episodes and the two-mode model in 29; the learned controller spills in 12, more than the slower pendulum controller (0.8) at their tuned margins, but less than the pendulum controller when both are run at the same transport time. When the ground truth is a single linear mode, the learned and pendulum controllers are indistinguishable. Sensing the liquid matters far more than which liquid sensor is used.

Index Terms—liquid sloshing, model predictive control, MPPI, learned dynamics, system identification, input shaping

I. Introduction

Carrying an open container that is filled to the brim is a task in which speed and safety pull in opposite directions: every acceleration of the container tilts and excites the free surface, and the surface has only a few millimetres of freeboard before liquid is lost. The practical objective, as stated in the brief for this study, is not merely to avoid spilling (that is easy if one moves slowly) but to be the fastest spill-free transport.

A controller that plans fast motions needs a prediction of the surface. The dominant choice is an equivalent mechanical model, a pendulum or a mass-spring-damper whose parameters are fitted to the liquid [1]–[3]. Such

models underlie slosh-free trajectory optimisation [4], feed-forward rest-to-rest manoeuvres [5], time-optimal planning with a bound on the surface height [6] and real-time MPC [7], [8]. A learned dynamics model is an alternative that has been used for spill-free transport [9]; it can absorb higher modes and nonlinear effects, and with a history encoder it can adapt to a liquid whose properties are not known in advance [10], [11].

What is missing is a like-for-like comparison: the same plant, optimiser, spill criterion and tuning rule, with only the slosh model exchanged. This paper reports such a comparison at small scale. The question is: does a learned slosh predictor inside sampling-based MPC enable faster spill-free transport than MPC with a fitted pendulum or spring-mass model, across liquid parameters and external pushes?

What we did. We built a planar cart-and-container simulation with a reduced-order two-mode nonlinear slosh ground truth, a time-minimising MPPI controller with a pluggable slosh model, three fitted parametric models (pendulum, spring-mass, two-mode spring-mass), a zero-vibration input-shaping baseline, and a learned predictor with and without latent context inference. Each system’s single safety parameter is tuned by one validation rule. We test four hypotheses fixed before the experiments:

- H1: on the nominal liquid the learned-model MPC is faster, spill-free, than the fitted-pendulum MPC.
- H2: on held-out liquids with pushes, the learned model with a history encoder spills less than the fixed-parameter pendulum MPC at equal transport time.
- H3: the learned advantage vanishes when the true slosh is a single linear mode.
- H4: the sensor set changes performance, and force/torque sensing is the most informative single addition to cart-only sensing.

What we found. H1 holds against the pendulum and the single-mode spring-mass model, but the learned controller is not the fastest system on the nominal liquid: a fitted two-mode spring-mass model and open-loop input shaping are faster there. H2 holds only in the equal-time sense; at each system’s own tuned margin the slower pendulum

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controller spills less. H3 holds. The first half of H4 holds and the second half is not supported. The study is a simulation with an invented reduced-order liquid, five seeds, and reimplemented baselines; Section VII states what this does and does not show.

II. Related Work

Equivalent mechanical slosh models. Sloshing is commonly represented for control design by equivalent mechanical systems, pendulums or mass-spring-dampers, one per mode [1]. Guagliumi et al. use a linear and a nonlinear mass-spring-damper model to estimate the sloshing height at the container wall for planar motions [2], and Di Leva et al. validate a sloshing-height model for two-dimensional motions of a cylindrical container experimentally [3]. Our parametric baselines are of this family, and our spill criterion is the same quantity, the surface height at the wall.

Slosh-aware planning and control. Muchacho et al. model the end effector as a point mass on a spherical pendulum and obtain slosh-free trajectories from a quadratic program [4]; Arrizabalaga et al. avoid the pendulum model with a geometric, virtual-quadrotor formulation for real-time slosh-free tracking [12]. Tóth et al. design fast rest-to-rest feedforward manoeuvres from multi-mode pendulum parameters identified for arbitrary container shapes [5]. Ferrari et al. plan time-optimal trajectories that keep the liquid surface below a threshold [6]; Medico et al. embed a nonlinear spherical-pendulum slosh model in real-time MPC for containers on a tray [7]; Hynninen and Kyrki formulate time-optimal spill-free emergency stopping as MPC [8]. These works fix the model class and study the planner; we fix the planner and vary the model class.

Learned models. Abderezaei et al. use learned dynamics for spill-free transport of open containers in clutter [9]. Bazzi et al. report that humans moving a cup of sloshing liquid appear to rely on simplified internal models [13], which is a reminder that a richer model is not automatically the better controller model. Learned models inside sampling-based planners are standard in model-based reinforcement learning, for example probabilistic ensembles with trajectory sampling [14], and Gaussian-process residual models have been combined with MPC [15].

Sampling-based MPC and adaptation. MPPI handles arbitrary dynamics and costs, and recent variants improve its sample efficiency or smoothness [16], [17]; our controller uses smooth, low-pass perturbations in that spirit. Inferring a latent description of the environment from recent history is the mechanism of rapid motor adaptation [10] and of implicit identification for dynamics models [11]; our history encoder applies it to liquid properties. Input shaping, the convolution of a command with an impulse sequence to cancel residual vibration [18], is our open-loop baseline.

III. Method

A. Plant and spill criterion

The cart is a double integrator with acceleration command u , $|u| \leq 4 \text{ m/s}^2$, updated every 0.03s. It carries a rectangular container of width $w = 0.12 \text{ m}$ whose rest surface is $f = 0.010 \text{ m}$ below the brim (the freeboard). The liquid is described by two antisymmetric modes with amplitudes p_1, p_2 in units of f . From linear modal theory for a rectangular tank, with $k_n = (2n - 1)\pi/w$ and liquid depth h ,

$$\omega_n^2 = g k_n \tanh(k_n h), \quad \gamma_n = \frac{4 \tanh(k_n h)}{w k_n f}, \quad (1)$$

and the ground truth used in the main experiments is

$$\ddot{p}_1 = -2\zeta\omega_1(1 + \delta|p_1|)\dot{p}_1 - \frac{\omega_1^2 p_1}{\sqrt{1 + \kappa p_1^2}} - \gamma_1 a, \quad (2)$$

$$\ddot{p}_2 = -2\zeta\omega_2\dot{p}_2 - \omega_2^2(p_2 + \epsilon p_1^3) - \gamma_2 a, \quad (3)$$

where a is the cart acceleration (command plus push) and $\zeta_2 = 1.5\zeta$. The surface elevations at the two walls are

$$s_{R,L} = \pm(p_1 + p_2) + \beta p_1^2, \quad (4)$$

so crests are higher than troughs are deep. An episode spills if $\max(s_R, s_L) > 1$ at any time. The nonlinear coefficients ($\kappa, \delta, \epsilon, \beta$; values in Section IV) are our own choice of plausible effects: softening stiffness, amplitude-dependent damping, energy transfer to the second mode and crest/trough asymmetry. They are not calibrated to a fluid simulation or to an experiment. The equations are integrated with a fourth-order Runge–Kutta scheme. The liquid properties that vary are the damping ratio ζ (a viscosity proxy) and the depth h (fill level), which shifts the frequencies.

B. Time-minimising MPPI

All model-based systems share one controller. At each step it samples acceleration sequences over a horizon of H steps around the current nominal sequence, rolls the cart forward exactly and the liquid forward with the plugged-in slosh model, and forms the cost

$$J = \Delta t \sum_{j=1}^H \rho(x_j, v_j) + T^*(x_H, v_H) + c \sum_{j=1}^H [\hat{s}_j - m]_+^2 + J_{\text{reg}}, \quad (5)$$

where ρ is a smooth indicator of not being at rest at the goal, T^* is the minimum time-to-go of a double integrator, $\hat{s}_j$ is the predicted larger wall elevation, m is the spill margin and J_{reg} contains a small penalty on acceleration and on acceleration increments plus a linear penalty on the largest violation. The nominal sequence is updated with the usual exponential weights [16]. Perturbations are piecewise-linear in time on a coarse knot grid, a low-pass sampling choice in the spirit of [17], and a few structured candidates (the nominal sequence retimed or rescaled) are added to the sample set. The margin m is the only quantity tuned per system.

C. Learned predictor with history encoder (Learned-Ctx)

The learned model has two multilayer perceptrons. The encoder reads the last K sensor readings and the last K measured cart accelerations and outputs a latent context z and an estimate of the last three frames of the wall elevations (s_R, s_L) . The dynamics network maps those three frames, the last two accelerations, the next acceleration and z to the increment of (s_R, s_L) . During an MPC rollout the encoder is evaluated once and z is held fixed, as in rapid motor adaptation [10]; the dynamics network is applied autoregressively. Both networks are trained jointly on excitation data from liquids drawn over a training range of (ζ, h) , with a loss that sums the frame-estimation error and the error of a multi-step unroll. The true wall elevations are available as training targets in simulation; at test time only the sensor readings are used. The no-context baseline uses the same encoder as a state estimator only, with no latent z .

D. Reimplemented baselines

Pendulum MPC: a single pendulum whose normal follows a flat liquid surface, $\ddot{\theta} = -(g/L)\sin\theta - c\dot{\theta} - (a/L)\cos\theta$, with wall elevation proportional to $\tan\theta$; three fitted parameters. Spring-mass MPC: a linear mass-spring-damper, three fitted parameters. Two-mode spring-mass MPC: two such modes in parallel, six fitted parameters. All three are fitted by output-error least squares on an identification experiment with the nominal liquid and obtain their state from a steady-state Kalman observer on the probe signal. ZV input shaping: a bang-bang acceleration profile convolved with a two-impulse zero-vibration shaper built from the fitted spring-mass frequency and damping, tracked by a PD loop on the cart; its tuned scalar is the peak acceleration. All baselines are reimplemented by us from their textbook form and are not the code of any cited paper. An oracle MPC that plans with the true model, state and parameters is reported as a privileged reference for the optimiser, not as a baseline.

IV. Experimental Setup

This is a small CPU-only study. The plant is integrated with our own numpy code rather than MuJoCo, which the brief proposed but which was not available in the compute environment; the cart is a double integrator, so nothing in the physics depends on that choice.

Liquids. Nominal: $\zeta = 0.03$, $h = 0.035$ m. Training range (learned model only): $\zeta \in [0.01, 0.08]$ log-uniform, $h \in [0.020, 0.050]$ m. Held-out: $\zeta \in [0.005, 0.01] \cup [0.08, 0.12]$ and $h \in [0.014, 0.020] \cup [0.050, 0.070]$ m, each side with equal probability. Nonlinear coefficients: $\kappa = 0.3$, $\delta = 0.5$, $\epsilon = 0.08$, $\beta = 0.15$.

Tasks. nominal: the nominal liquid. push: the nominal liquid and one lateral push that changes the cart velocity by 0.04 m/s at a random time. heldout: held-out liquids. heldout_push: held-out liquids with the same push. lin1: the nominal liquid with a single linear mode as ground

TABLE I

Tuned scalar per system from the validation rule: spill margin m for MPC systems, peak acceleration in m/s^2 for ZV shaping.

System	nominal truth	single linear mode
Learned-Ctx MPC	0.90	0.95
Learned MPC (no context)	0.90	0.95
Pendulum MPC	0.70	0.95
Spring-mass MPC	0.80	1.00
Two-mode spring-mass MPC	0.85	1.00
ZV input shaping	1.40	1.90

truth (all nonlinear coefficients and γ_2 set to zero); every model is refitted or retrained on that truth. An episode lasts 3.3 s; the cart has arrived when it stays within 0.01 m of the goal with speed below 0.03 m/s.

Sensors. Cart position, velocity and measured acceleration are always available. The default liquid sensor is a slosh-height probe at both walls with noise of standard deviation 0.03 freeboards. Alternatives in the sensor ablation: a force/torque proxy (the specific lateral force of the liquid, noise 0.05 m/s^2), a camera proxy (surface tilt with noise 0.08 and a latency of 2 steps), and no liquid sensor.

Data and training. The learned model uses 50000 transitions (500 trajectories of 100 steps) of random, chirp and transport-like excitation over the training range, with $K = 30$, a 4-dimensional context, hidden widths 128 and 96, a 15-step unroll and 2500 Adam steps. The parametric models use 40 trajectories of 100 steps of the nominal liquid. Because a brim-full container cannot be excited far beyond the brim, excitation trajectories are redrawn at lower amplitude if the surface exceeds 1.5 freeboards.

Controller. MPPI with 64 samples, $H = 30$ steps, 2 iterations per control step, shared by all model-based systems. These settings were chosen once with the oracle model on a development seed that is not reported.

Tuning protocol. For each system the tuned scalar is the largest grid value such that it and all smaller grid values produce no spill and full arrival on validation episodes of the nominal task (3 seeds, 30 episodes each). The lin1 task has its own selection. Test episodes use a different random stream; no test episode enters any choice. The selected values are listed in Table I.

Seeds and metrics. Main results use 5 seeds; a seed fixes the identification and training data, the network initialisation and the episodes. Each run has 50 test episodes. Ablations and sweeps use 3 seeds. We report the mean transport time (an episode that never arrives counts as the episode length), the spill rate, the mean per-episode peak wall elevation in freeboards, and the open-loop prediction error of the wall elevations 20 steps ahead on fresh excitation trajectories of the task's liquids. Tables give mean $\pm$ standard deviation over seeds, and comparisons use Welch's t -test on per-seed values. The experiments took about 39 minutes on two CPU threads.

V. Results

Table II gives the main results, Fig. 1 the trade-off between transport time and spill rate as the tuned scalar is varied, and Fig. 2 one example episode per system.

A. H1: nominal liquid

All systems are spill-free on the nominal test episodes at their tuned settings. The learned-context controller needs 1.993s and the pendulum controller 2.147s, a change of -7.2 relative to the pendulum (Welch $p = 2.448 \times 10^{-8}$); the per-seed ranges do not overlap. H1 is supported. Three qualifications matter more than the headline.

First, the size of the gap depends on the margin grid. The pendulum and the single-mode spring-mass model have almost the same prediction error (Table II), yet the validation rule gives them margins of 0.70 and 0.80, because the pendulum controller spilled in one validation run at the larger value. Against the spring-mass controller (2.041s) the change is -2.3 .

Second, the learned model is not the fastest model-based system. The two-mode spring-mass controller is spill-free at 1.965s, slightly faster than the learned controller (Welch $p = 0.001148$). The learned model has the lower 20-step prediction error on the nominal liquid and runs at a larger margin (Table I), but that did not translate into a shorter time.

Third, open-loop ZV input shaping is the fastest system on the nominal liquid, 1.890s, and it is also faster than the privileged oracle MPC (1.916s, spill rate 0.8 at margin 1.00). Two things explain this. The sampling optimiser is not exactly time-optimal: with a perfect model it still leaves time on the table. And shaping, being deterministic on a fixed liquid, can be tuned to a peak elevation of 0.975 freeboards with no validation spill. The model-related differences among MPC systems (a few hundredths of a second) are therefore of the same order as the loss due to the optimiser.

B. H2: held-out liquids and pushes

A push on the nominal liquid already separates the systems (task push in Table II): input shaping spills in 99 of episodes, the learned-context controller in 2.4, the two-mode spring-mass controller in 5.6 and the pendulum controller in none.

The picture reverses for input shaping as soon as the liquid differs from the one it was tuned on: it spills in 100 of held-out episodes. Among MPC systems at their tuned margins on heldout_push, the learned-context controller takes 2.024s with a spill rate of 12.0; the pendulum controller takes 2.270s with 0.8. So at the tuned margins the pendulum controller spills less (Welch $p = 0.0007056$); it buys that with a longer transport: the difference in mean time, learned minus pendulum, is -0.25 s. Margins tuned on the nominal liquid do not transfer for the learned model: a margin with no nominal validation spill gives a

non-zero spill rate on held-out liquids even without a push (2.4).

H2 is stated at equal transport time, which the right panel of Fig. 1 addresses. At a margin of 0.8 both controllers take about the same time (2.129s learned, 2.128s pendulum) and the spill rates are 0.7 and 18.0. Conversely, the pendulum controller reaches a spill rate of 0.7 only at 2.271s. H2 is supported in the equal-time sense (3 seeds), and not in the sense of a lower spill rate at the deployed settings. The context matters here: the no-context learned controller has a higher 20-step prediction error on held-out liquids (Table II, $p = 1.232 \times 10^{-7}$) and its curve in Fig. 1 lies between the other two; its spill-rate difference to the context model at the tuned margin is not significant on heldout_push ($p = 0.2066$).

The two-mode spring-mass controller, the fastest MPC system on the nominal liquid, is the least robust of them: 29.2 spills on heldout_push. Its extra accuracy is specific to the liquid it was identified on.

C. H3: single linear mode

With a single linear mode as ground truth the fitted models are structurally exact, and their 20-step error is lower than that of the learned model (Table II). The learned-context and pendulum controllers take 1.797s and 1.806s (Welch $p = 0.2306$), both without spills. H3 is supported: the advantage over the pendulum is within seed noise. Input shaping is again fastest (1.650s); its tuned peak acceleration is the largest value of the grid we searched, so it might be faster still.

VI. Ablations and Analysis

All ablations use the heldout_push task, the learned-context controller at its tuned margin, 3 seeds and 50 episodes per run. With 3 seeds, only large differences can be resolved.

H4: sensors (Table III). Without any liquid sensor the spill rate is 67, against 12 with the probe ($p = 0.02621$): on an unknown liquid, the action history alone does not identify the liquid. Adding force/torque gives 15 and adding the camera proxy 25. Force/torque is close to the probe and their difference is not significant ($p = 0.5529$); the probe has the lowest mean. The first part of H4 is supported; the claim that force/torque is the most informative single addition is not. The ranking of the three liquid sensors also depends on the noise levels we assigned, which were not swept, so the “per unit noise” part of H4 is untested.

History length (Table IV). Removing the encoder altogether roughly doubles the 20-step prediction error and lengthens the transport, because the dynamics network then works from raw noisy frames. Ten steps of history give a higher prediction error than thirty, but the closed-loop spill rates of the two are indistinguishable.

Data size (Table V). With 1k transitions the controller is unreliable; 10k transitions are close to 50k in prediction error and within noise in spill rate.

TABLE II

Main results at each system’s tuned setting (mean $\pm$ std over 5 seeds, 50 test episodes per seed). Columns: mean transport time [s], spill rate, mean peak wall elevation [freeboards], 20-step prediction RMSE [freeboards]. Tasks: nominal liquid, nominal liquid with a push, held-out liquids, held-out liquids with a push, single-linear-mode ground truth. Bold and underline mark the best and second-best value of each column separately; a low transport time is only meaningful together with the spill rate.

Method	Task	transport_time_s $\downarrow$	spill_rate $\downarrow$	peak_disp $\downarrow$	pred_rmse_20 $\downarrow$	n
Pendulum MPC	nominal	2.147 $\pm$ 0.012	0.000 $\pm$ 0.000	0.799 $\pm$ 0.010	0.135 $\pm$ 0.005	5
ZV input shaping	nominal	1.890 $\pm$ 0.000	0.000 $\pm$ 0.000	0.975 $\pm$ 0.007	–	5
Learned MPC (no context)	nominal	1.987 $\pm$ 0.018	0.000 $\pm$ 0.000	0.904 $\pm$ 0.022	0.092 $\pm$ 0.003	5
Spring-mass MPC	nominal	2.041 $\pm$ 0.014	0.000 $\pm$ 0.000	0.925 $\pm$ 0.011	0.136 $\pm$ 0.005	5
Two-mode spring-mass MPC	nominal	1.965 $\pm$ 0.007	0.000 $\pm$ 0.000	0.904 $\pm$ 0.007	0.094 $\pm$ 0.007	5
Learned-Ctx MPC	nominal	1.993 $\pm$ 0.010	0.000 $\pm$ 0.000	<u>0.887 $\pm$ 0.013</u>	0.063 $\pm$ 0.004	5
Pendulum MPC	heldout	2.193 $\pm$ 0.008	0.000 $\pm$ 0.000	0.800 $\pm$ 0.014	0.473 $\pm$ 0.015	5
ZV input shaping	heldout	1.890 $\pm$ 0.000	1.000 $\pm$ 0.000	1.253 $\pm$ 0.021	–	5
Learned MPC (no context)	heldout	2.043 $\pm$ 0.024	0.100 $\pm$ 0.071	0.902 $\pm$ 0.033	0.316 $\pm$ 0.013	5
Spring-mass MPC	heldout	2.085 $\pm$ 0.010	0.104 $\pm$ 0.065	0.920 $\pm$ 0.016	0.470 $\pm$ 0.014	5
Two-mode spring-mass MPC	heldout	2.009 $\pm$ 0.015	0.228 $\pm$ 0.101	0.961 $\pm$ 0.021	0.462 $\pm$ 0.013	5
Learned-Ctx MPC	heldout	<u>1.996 $\pm$ 0.009</u>	<u>0.024 $\pm$ 0.017</u>	<u>0.892 $\pm$ 0.019</u>	0.176 $\pm$ 0.012	5
Pendulum MPC	heldout_push	2.270 $\pm$ 0.032	0.008 $\pm$ 0.011	0.812 $\pm$ 0.014	0.473 $\pm$ 0.015	5
ZV input shaping	heldout_push	1.890 $\pm$ 0.000	0.972 $\pm$ 0.023	1.276 $\pm$ 0.026	–	5
Learned MPC (no context)	heldout_push	2.057 $\pm$ 0.008	0.184 $\pm$ 0.093	0.922 $\pm$ 0.026	<u>0.316 $\pm$ 0.013</u>	5
Spring-mass MPC	heldout_push	2.160 $\pm$ 0.049	0.132 $\pm$ 0.097	0.932 $\pm$ 0.018	0.470 $\pm$ 0.014	5
Two-mode spring-mass MPC	heldout_push	2.029 $\pm$ 0.041	0.292 $\pm$ 0.125	0.972 $\pm$ 0.025	0.462 $\pm$ 0.013	5
Learned-Ctx MPC	heldout_push	<u>2.024 $\pm$ 0.006</u>	<u>0.120 $\pm$ 0.032</u>	<u>0.915 $\pm$ 0.011</u>	0.176 $\pm$ 0.012	5
Pendulum MPC	lin1	1.806 $\pm$ 0.005	0.000 $\pm$ 0.000	0.898 $\pm$ 0.003	0.033 $\pm$ 0.000	5
ZV input shaping	lin1	1.650 $\pm$ 0.000	0.000 $\pm$ 0.000	0.949 $\pm$ 0.000	–	5
Learned MPC (no context)	lin1	1.799 $\pm$ 0.009	0.004 $\pm$ 0.009	0.933 $\pm$ 0.010	0.066 $\pm$ 0.010	5
Spring-mass MPC	lin1	<u>1.775 $\pm$ 0.006</u>	0.012 $\pm$ 0.018	0.944 $\pm$ 0.005	0.032 $\pm$ 0.001	5
Two-mode spring-mass MPC	lin1	<u>1.777 $\pm$ 0.005</u>	0.016 $\pm$ 0.009	0.944 $\pm$ 0.002	<u>0.032 $\pm$ 0.001</u>	5
Learned-Ctx MPC	lin1	1.797 $\pm$ 0.014	0.000 $\pm$ 0.000	<u>0.929 $\pm$ 0.007</u>	0.046 $\pm$ 0.003	5
Pendulum MPC	push	2.183 $\pm$ 0.014	0.000 $\pm$ 0.000	0.813 $\pm$ 0.013	0.135 $\pm$ 0.005	5
ZV input shaping	push	1.890 $\pm$ 0.000	0.988 $\pm$ 0.018	1.090 $\pm$ 0.008	–	5
Learned MPC (no context)	push	2.002 $\pm$ 0.023	0.028 $\pm$ 0.033	0.918 $\pm$ 0.017	<u>0.092 $\pm$ 0.003</u>	5
Spring-mass MPC	push	2.088 $\pm$ 0.032	0.072 $\pm$ 0.076	0.938 $\pm$ 0.013	0.136 $\pm$ 0.005	5
Two-mode spring-mass MPC	push	<u>1.981 $\pm$ 0.010</u>	0.056 $\pm$ 0.026	0.922 $\pm$ 0.005	0.094 $\pm$ 0.007	5
Learned-Ctx MPC	push	<u>2.012 $\pm$ 0.007</u>	<u>0.024 $\pm$ 0.017</u>	<u>0.903 $\pm$ 0.008</u>	0.063 $\pm$ 0.004	5

TABLE III

Sensor set (learned-context controller, heldout_push). The row “Learned-Ctx MPC” is the default, cart + slosh-height probe. Columns as in Table II.

Method	transport_time_s $\downarrow$	spill_rate $\downarrow$	pred_rmse_20 $\downarrow$	n
cart only	2.055 $\pm$ 0.042	0.673 $\pm$ 0.163	0.381 $\pm$ 0.017	3
cart + camera proxy	2.013 $\pm$ 0.010	0.247 $\pm$ 0.099	0.232 $\pm$ 0.007	3
cart + force/torque	2.022 $\pm$ 0.016	0.153 $\pm$ 0.081	0.177 $\pm$ 0.007	3
Learned-Ctx MPC	2.026 $\pm$ 0.007	0.120 $\pm$ 0.020	0.172 $\pm$ 0.014	3

TABLE IV

Encoder history length. The default uses 30 steps; “history 0” feeds the last three raw probe frames to the dynamics network and has no latent context.

Method	transport_time_s $\downarrow$	spill_rate $\downarrow$	pred_rmse_20 $\downarrow$	n
history 10	2.030 $\pm$ 0.022	0.113 $\pm$ 0.012	0.238 $\pm$ 0.007	3
history 0 (no encoder)	2.241 $\pm$ 0.087	0.160 $\pm$ 0.072	0.373 $\pm$ 0.008	3
Learned-Ctx MPC	2.026 $\pm$ 0.007	<u>0.120 $\pm$ 0.020</u>	0.172 $\pm$ 0.014	3

Controller (Table VI). Horizon and sample count change the transport time by amounts comparable to the differences between models in Table II: a shorter horizon and more samples were faster on this task, a longer horizon and fewer samples slower. Our default was fixed beforehand with the oracle on the nominal liquid and was not re-tuned; this confirms that the optimiser is a first-order

TABLE V

Training-set size of the learned predictor (default 50k transitions).

Method	transport_time_s $\downarrow$	spill_rate $\downarrow$	pred_rmse_20 $\downarrow$	n
10k transitions	<u>2.054 $\pm$ 0.021</u>	<u>0.167 $\pm$ 0.064</u>	<u>0.194 $\pm$ 0.031</u>	3
1k transitions	2.356 $\pm$ 0.251	0.500 $\pm$ 0.173	0.397 $\pm$ 0.054	3
Learned-Ctx MPC	2.026 $\pm$ 0.007	0.120 $\pm$ 0.020	0.172 $\pm$ 0.014	3

TABLE VI

MPPI horizon and number of samples (default 30 steps, 64 samples). The model is the same in every row.

Method	transport_time_s $\downarrow$	spill_rate $\downarrow$	pred_rmse_20 $\downarrow$	n
horizon 45	2.171 $\pm$ 0.025	0.087 $\pm$ 0.031	0.172 $\pm$ 0.014	3
32 samples	2.158 $\pm$ 0.008	0.093 $\pm$ 0.064	0.172 $\pm$ 0.014	3
128 samples	<u>1.992 $\pm$ 0.031</u>	0.113 $\pm$ 0.081	0.172 $\pm$ 0.014	3
horizon 15	<u>1.963 $\pm$ 0.021</u>	0.113 $\pm$ 0.050	<u>0.172 $\pm$ 0.014</u>	3
Learned-Ctx MPC	2.026 $\pm$ 0.007	0.120 $\pm$ 0.020	0.172 $\pm$ 0.014	3

factor in the comparison.

Push magnitude (Fig. 3). On the nominal liquid, input shaping loses its advantage with the smallest push we tested, since the PD loop that rejects the push re-excites the liquid without regard to the surface. The MPC systems tolerate pushes up to 0.04 m/s with few spills; at 0.08 m/s the spill rates are 23 (learned-context), 29 (two-mode spring-mass) and 11 (pendulum). The pendulum

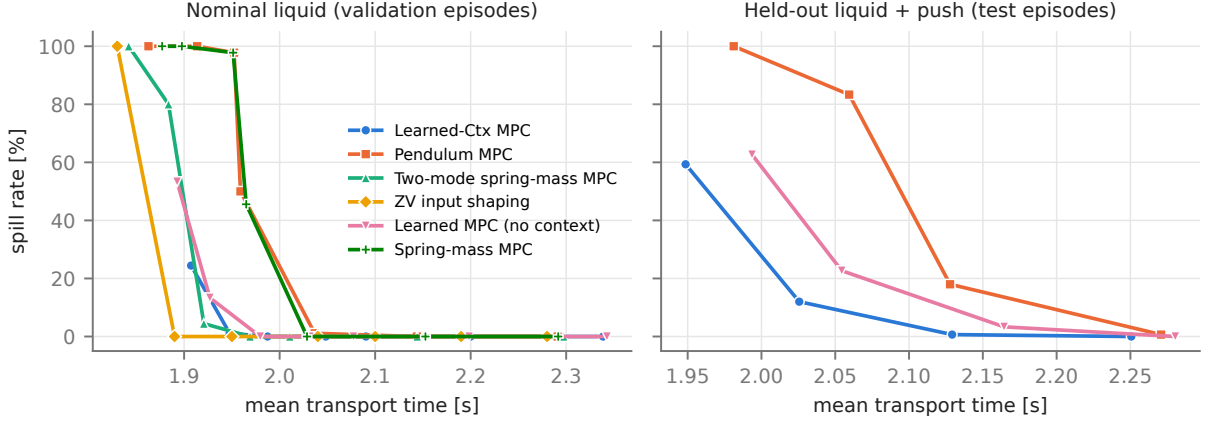


Fig. 1. Spill rate against mean transport time as each system’s tuned scalar is swept (mean over 3 seeds). Left: nominal liquid, validation episodes used for tuning. Right: held-out liquids with a push, test episodes, three systems. Curves closer to the lower left are better.

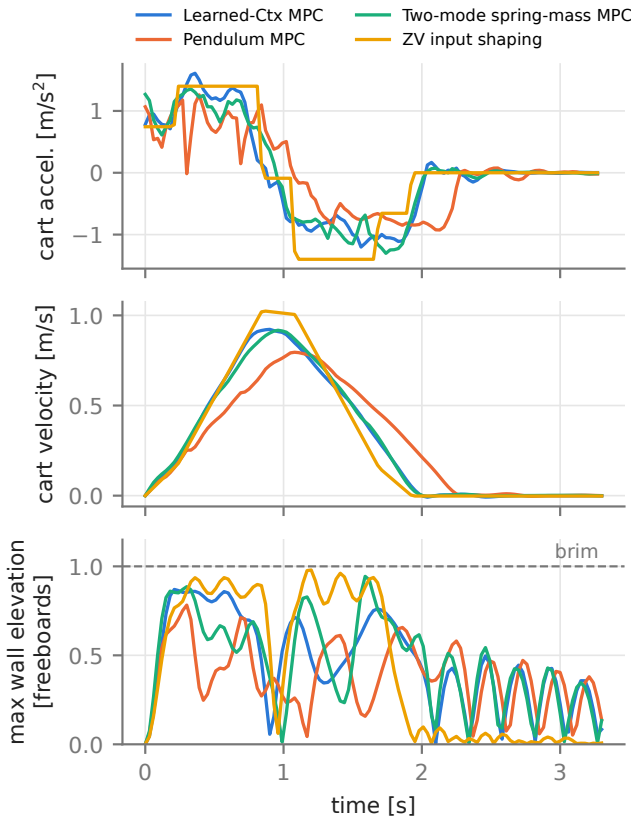


Fig. 2. One test episode on the nominal liquid (seed 0) for four systems at their tuned settings: commanded acceleration, cart velocity and the larger wall elevation.

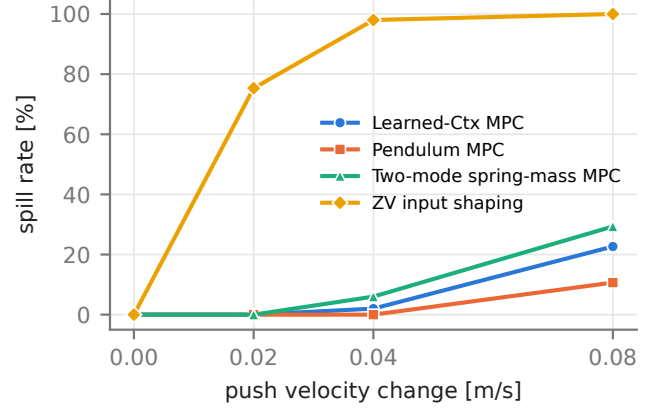


Fig. 3. Spill rate against push magnitude on the nominal liquid (3 seeds), each system at its tuned setting.

VII. Limitations

The liquid is a reduced-order model that we wrote. The ground truth is two modes with hand-chosen nonlinear terms, not a Navier–Stokes or particle simulation and not a real liquid. A learned model facing real sloshing meets breaking waves, three-dimensional modes and splashing; a pendulum model meets the same. Whether the ordering reported here survives is unknown. The nonlinearity strengths decide how wrong a single-mode model is, and we tested one setting plus the single-linear-mode extreme; we did not test a two-linear-mode truth, which would separate the effect of the second mode from that of the nonlinearity.

Scale. One planar axis, one container geometry, one transport distance, 5 seeds for the main table and 3 for ablations and sweeps, 50 episodes per run. Spill rates near zero are resolved only coarsely. The brief asked for 100 episodes per condition and seed and for a recurrent

controller, which runs with the most conservative margin, has the most room to absorb a push.

encoder; we used 250 episodes per condition pooled over seeds and a feedforward encoder to stay within the compute budget.

The optimiser is not time-optimal. The oracle MPC is slower than tuned open-loop input shaping on the nominal liquid, and the controller ablation moves the transport time by as much as the model choice does. Differences of a few hundredths of a second between MPC systems should not be read as properties of the models alone. A gradient-based or exact time-optimal solver would give a cleaner comparison.

Tuning rule and grid. Each system has one tuned scalar chosen on a grid from 90 validation episodes of the nominal liquid. The pendulum and spring-mass models, which predict almost equally well, received different margins from this rule; part of the reported gap between the learned and the pendulum controller is therefore a selection effect. Margins were not re-tuned for held-out liquids.

Baselines. All baselines are our reimplementations of textbook forms. The parametric models have fixed parameters identified on the nominal liquid; an adaptive pendulum with online parameter estimation, a robust input shaper, or a learned residual on a pendulum model would be stronger baselines on held-out liquids and were not tested. The learned model saw a range of liquids in training while the parametric models saw one, which favours the learned model on held-out liquids and the parametric models on the nominal liquid.

Sensing. The sensors are simulated proxies with noise levels we assigned; the camera is a noisy, delayed tilt signal, not an image pipeline. The learned model is trained with true surface heights as targets, which a real system would have to obtain from instrumentation. Grasp selection, arm kinematics, tilting the container and spill detection with stopping, all part of the original seed, are outside this study.

VIII. Conclusion

In this small simulation study, replacing a fitted single-mode pendulum or spring-mass model by a learned slosh predictor inside the same time-minimising MPPI controller shortened spill-free transport of a brim-full container on the nominal liquid: the pendulum controller took 1.077 times as long, with part of that gap attributable to the margin selection. The learned model was not the best option everywhere: a fitted two-mode spring-mass model was slightly faster on the liquid it was identified on, open-loop input shaping was fastest of all there, and at their deployed margins the slower pendulum controller spilled less on held-out liquids with pushes. What the learned model with a history encoder did provide was the better time-spill trade-off on liquids it had not been tuned for, where input shaping and the two-mode model failed. With a single linear slosh mode the learned model offered nothing over the pendulum. A liquid sensor of any of the

three kinds was far more important than the choice among them.

For a system that must be the fastest spill-free transporter, these results suggest three next steps: a solver that is closer to time-optimal, since the optimiser cost us as much as the model; a margin that adapts to the inferred liquid rather than one tuned on a nominal liquid; and a test against a fluid simulation or a real container, where the open question is whether the effects a learned model can absorb are large enough to matter.

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